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2025

BUPR – ERP

- 7 lectures
- Theory
- Practical assignment?

So... ERP

- What is it?



Business Process Management - ERP

- Business Process

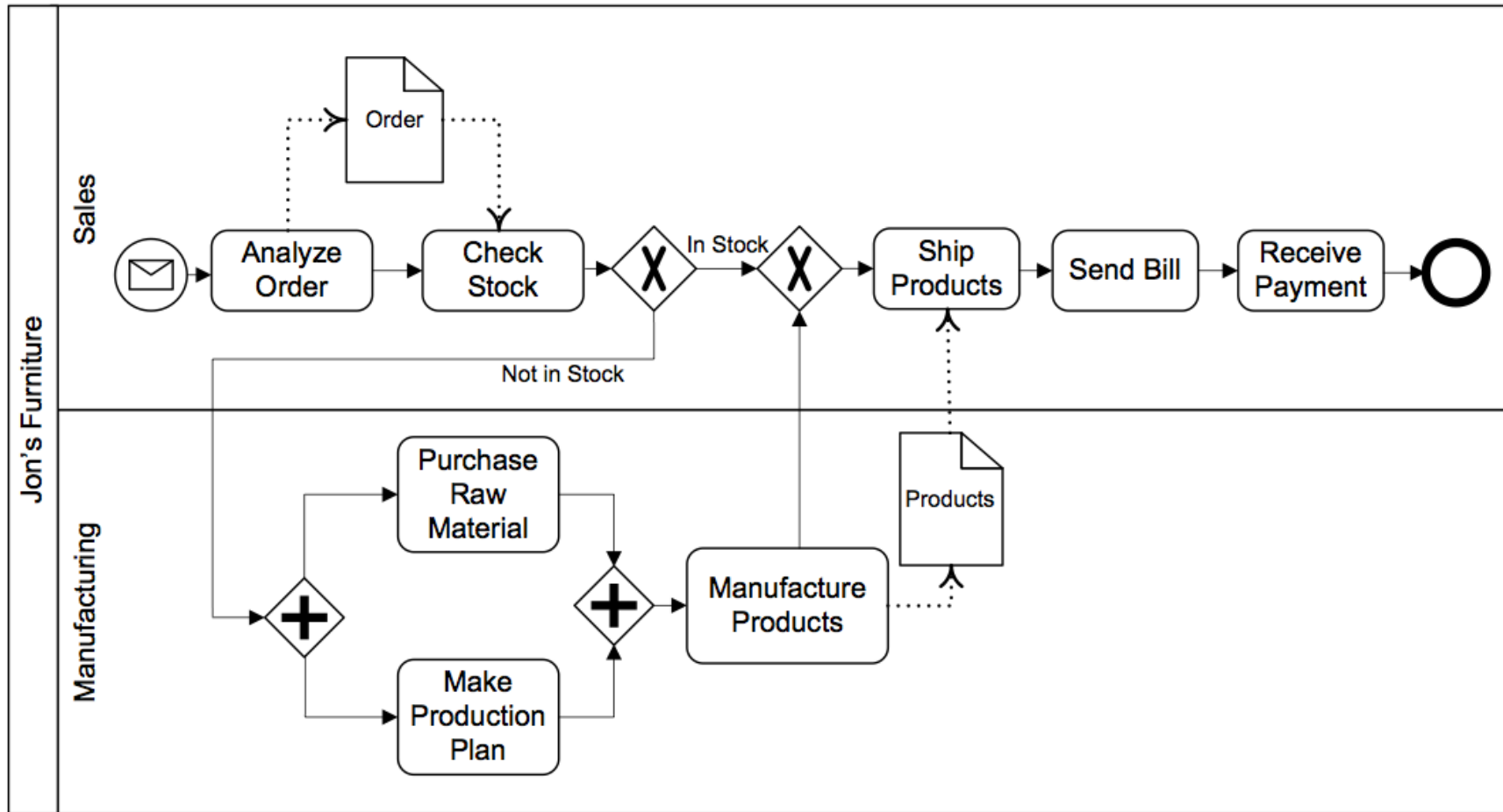
consists of a set of activities that are performed in **coordination** in an organizational and technical environment. These activities jointly realize a **business goal**. Each business process is enacted by a single organization, but it may interact with business processes performed by other organisations.

- Business Process Management

includes concepts, methods and techniques **to support the design, administration, configuration, enactment, and analysis** of business processes.

- Business Process Management System

is a generic software system that is driven by **explicit process representations** to coordinate the enactment of business processes.



M. Weske: Business Process Management,
© Springer-Verlag Berlin Heidelberg 2012, 2007

Fig. 4.78. Business process diagram with role information

Agenda

- ERP definition
- Historical development
- Major characteristics of an ERP system
- Importance for business

Enterprise Resource Planning

ENTERPRISE RESOURCE PLANNING integrates logistics, financial and human resources in **one system**.

Schenk, D., Draijer, C. and Caris, A. (2008). Hands-on with SAP ERP and IDES. 1st ed. Harlow: Pearson Education Limited, p.7.

ENTERPRISE RESOURCE PLANNING denotes the entrepreneurial task of

- **efficiently** making use of the **resources available** in/for the organization for operations
- And thus, **optimizing** the business **processes**.

Translated from: De.wikipedia.org. (2017). Enterprise-Resource-Planning.

ENTERPRISE RESOURCE PLANNING is **software** that facilitates the **flow of information** between **all functions** of a business, such as production, logistics, finance and personnel & organisation.

Visser, H. and Van Goor, A. (2011). Logistics: principles and practice. 2nd ed. 's-Gravendeel: Hessel Visser, p.67.

ENTERPRISE RESOURCE PLANNING systems focus primarily on **intra-company processes** – that is, the operations that are performed within an organization – and they **integrate** functional and cross-functional **business processes**.

*Magal, S. and Word, J. ed., (2012). Introduction to Business Process.
In: Integrated Business Processes with ERP Systems, 1st ed. Hoboken: Wiley, p.25.*



ERP

ERP suppliers

- Worldwide:

Oracle

SAP

Microsoft

INFOR

- The Netherlands:

AFAS

EXACT

ORACLE®

ENTERPRISE
RESOURCE PLANNING
CLOUD



Microsoft
Dynamics®



AFAS
software

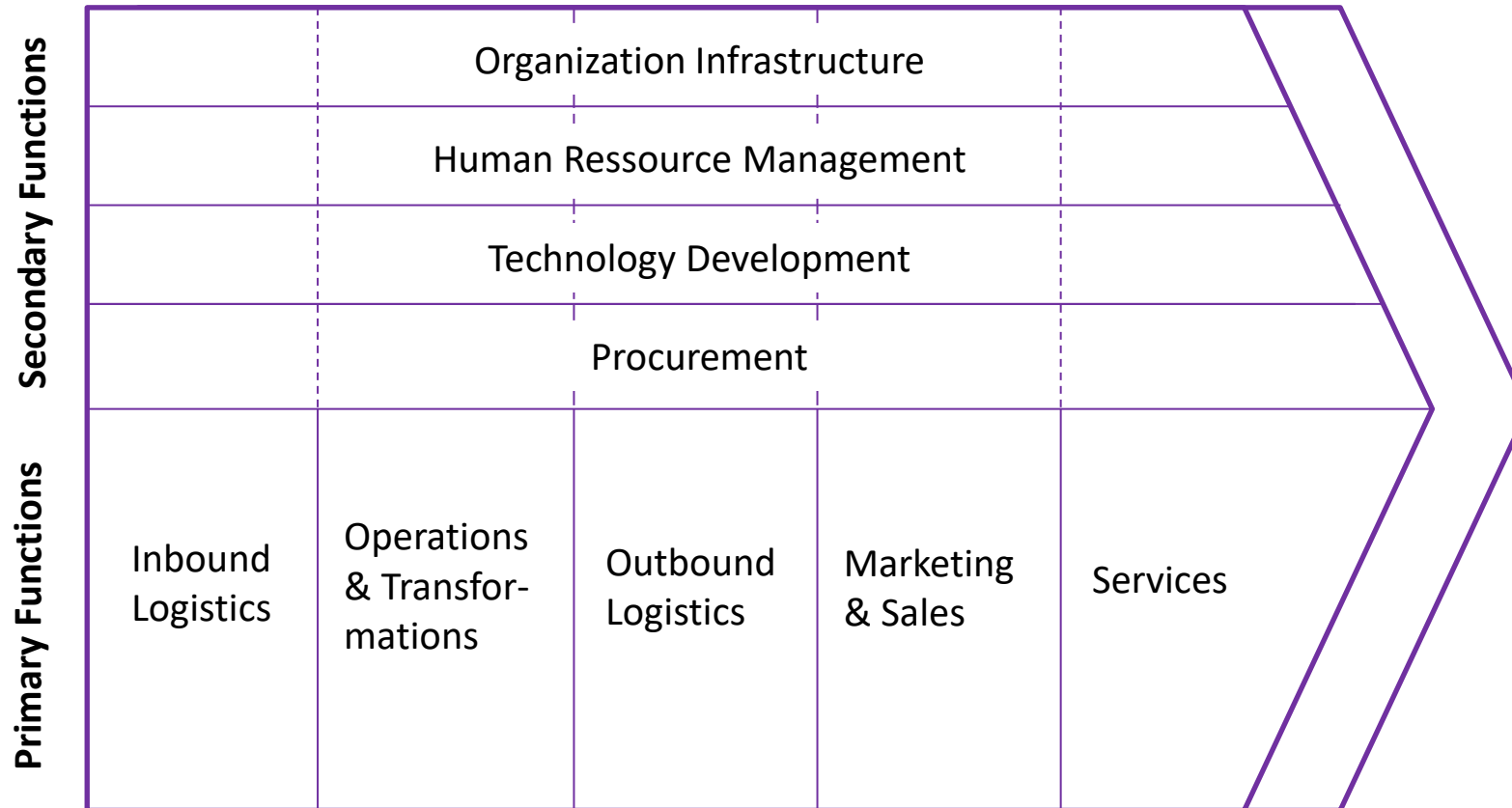
= EXACT

Overview of ERP, history and evolution

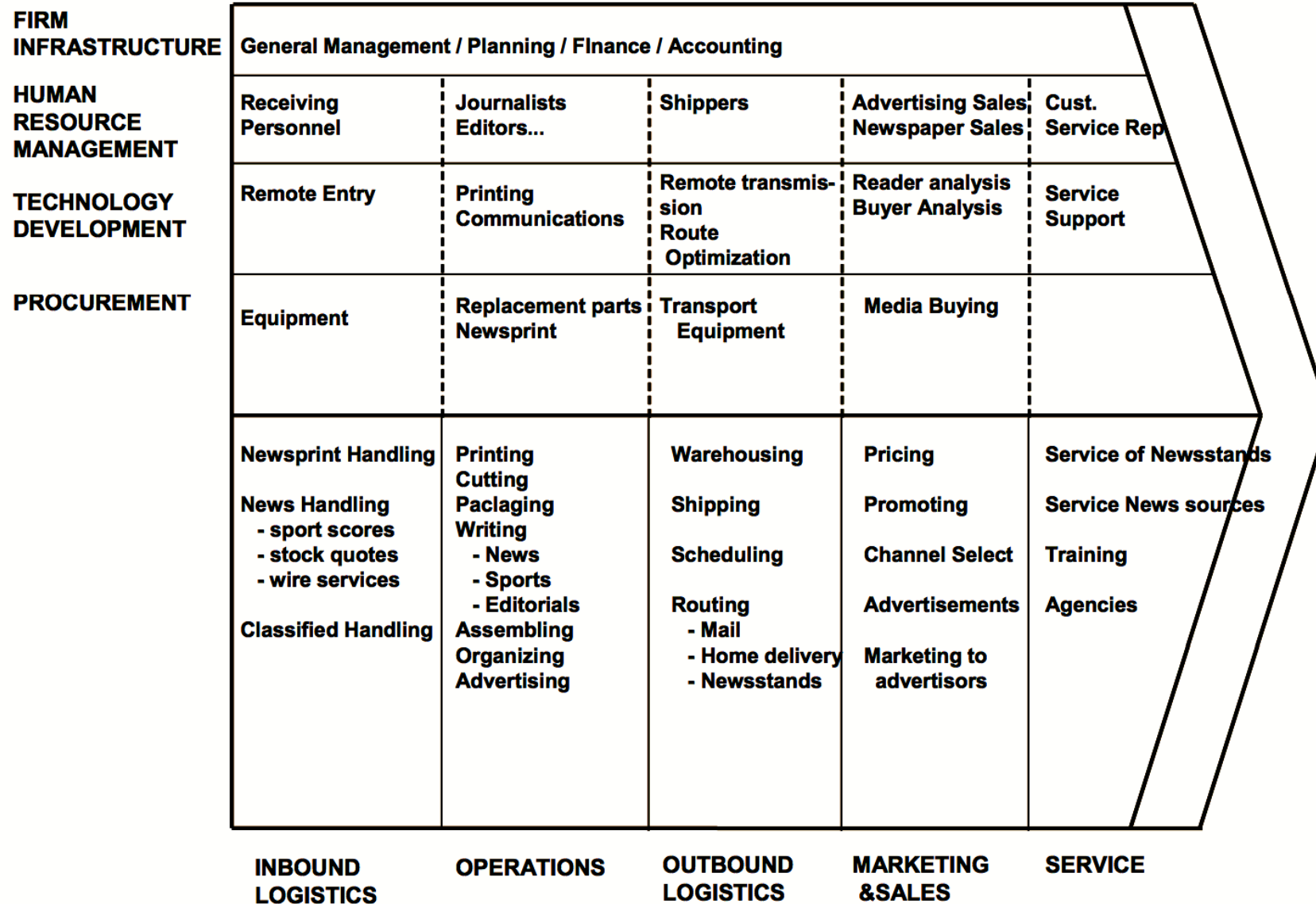
- Pre-ERP Era: independent systems (legacy systems) for **individual** business functions (finance, inventory, HR, etc.).
- Evolution:
 - Material Requirements Planning (MRP) 1960's-1970's
 - + Production planning
 - Manufacturing Resource Planning (MRP II) 1980's
 - + Finance and HRM
 - Full-scale ERP systems

Traditionally, organizations were functionally structured

(aka **VALUE CHAIN**, defined by
Michael Porter 1985)



Example for a Newspaper Enterprise



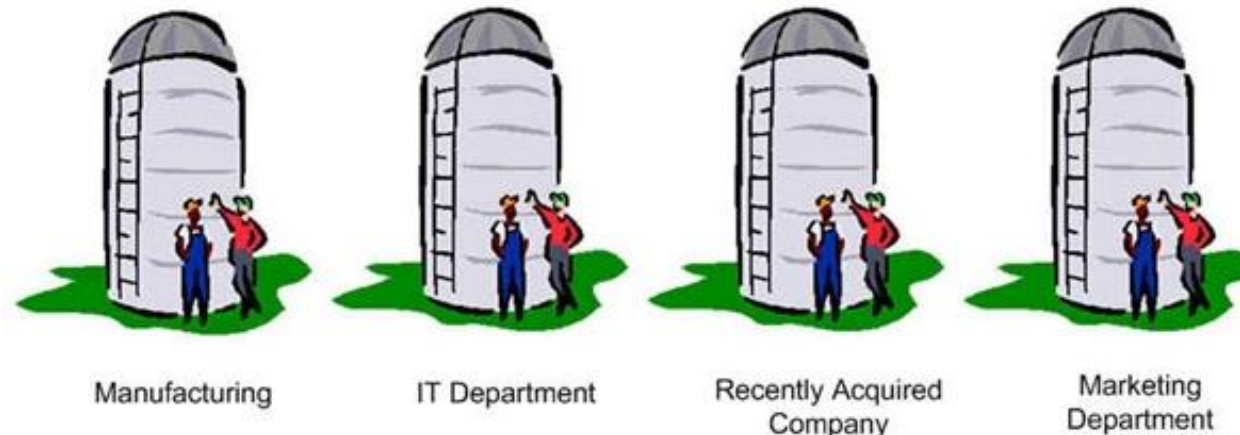
Problems with the functional structure

In phases of growth:

- Advantage to concentrate on “one” function.
- Isolated, function-oriented IT-systems built and used within functional areas only, e.g., software for material resource planning (MRP), ..

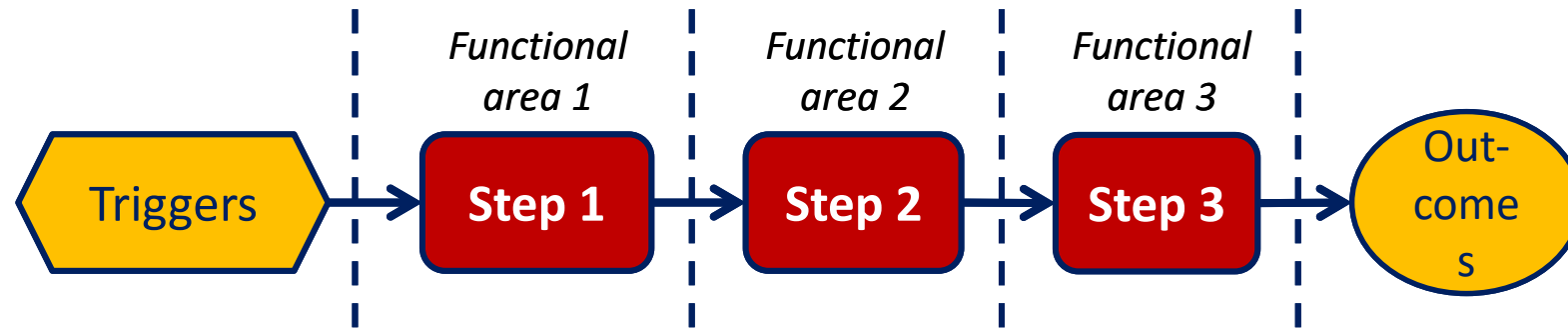
Over time, however, this results in the so-called **SILO EFFECT**:

- People perform their tasks in isolation, without understanding which steps happen before and which next.
 - They’ve lost the “big picture”, did not take overall responsibility anymore.
- The company loses efficiency and competitiveness.

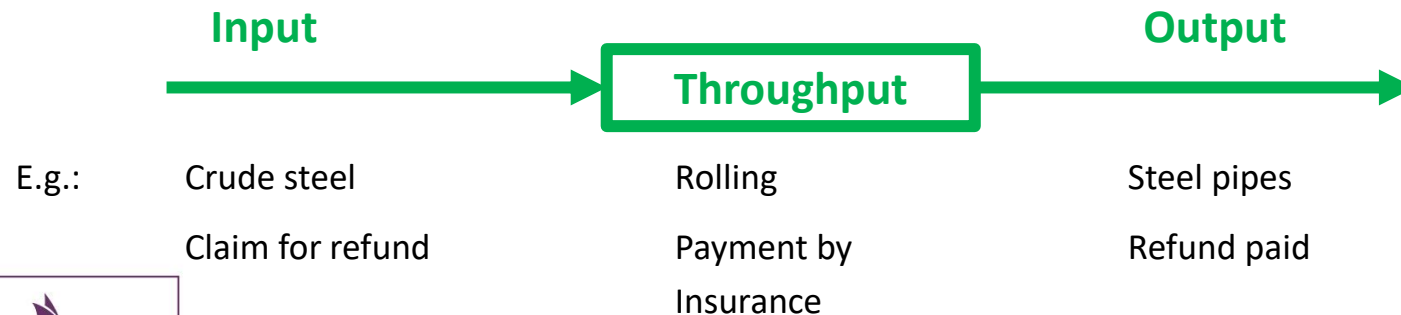


Business Process

A *Business Process* is a set of tasks / activities which produce desired outcomes. Every process is triggered by some event.

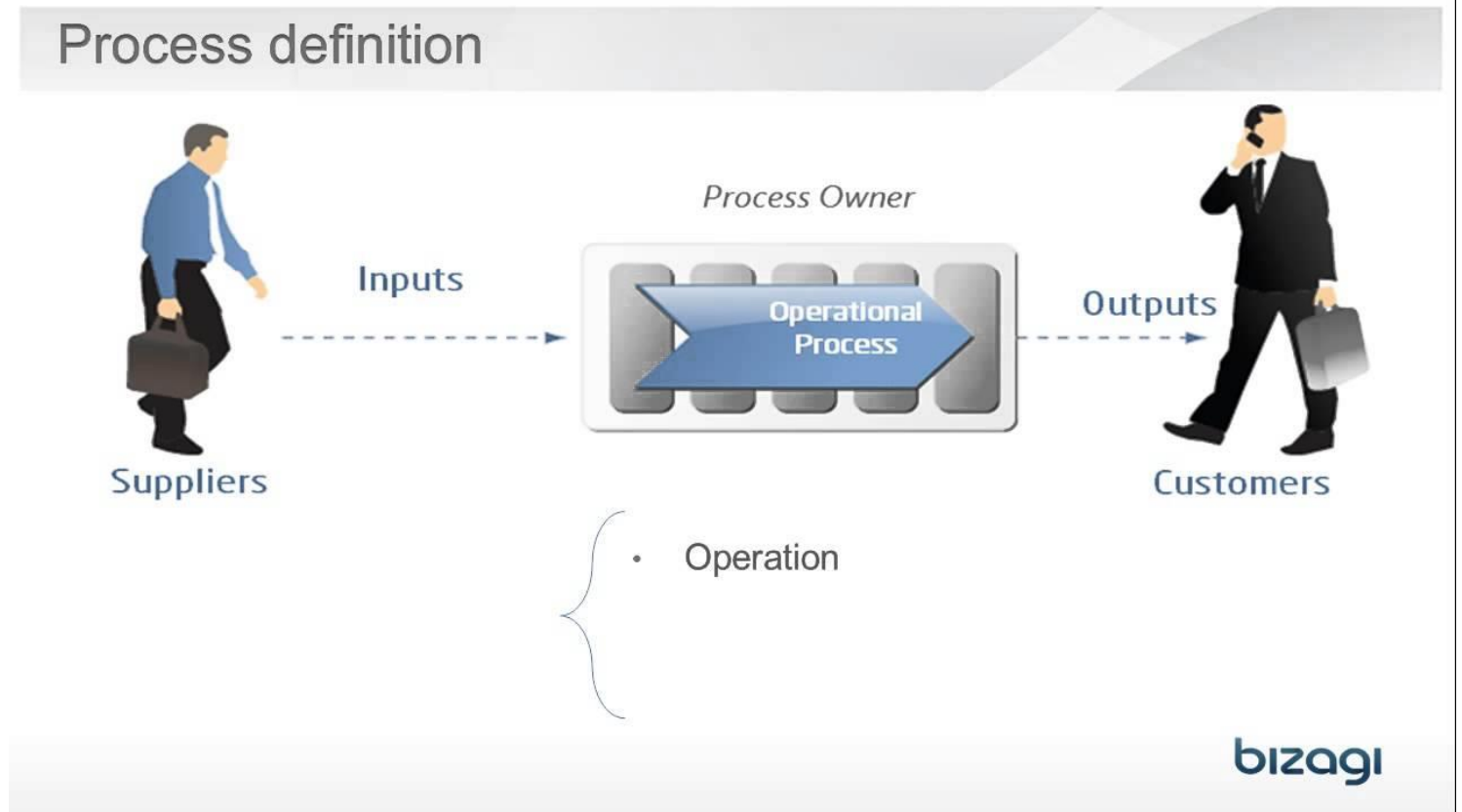


All organizations basically focus on making their goods more valuable. This means that all their physical processes can, in effect, be reduced to Value Adding.

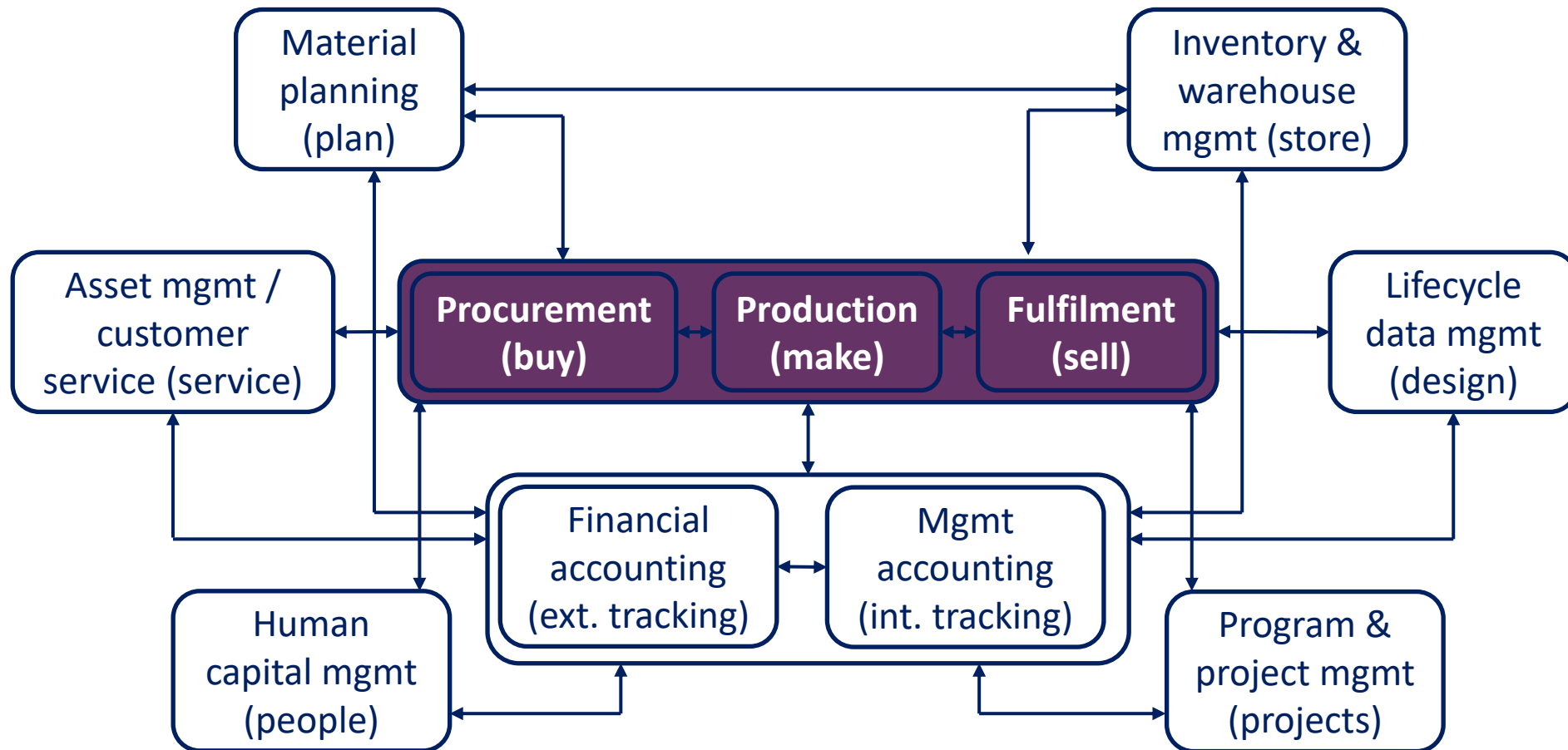


Examples of processes

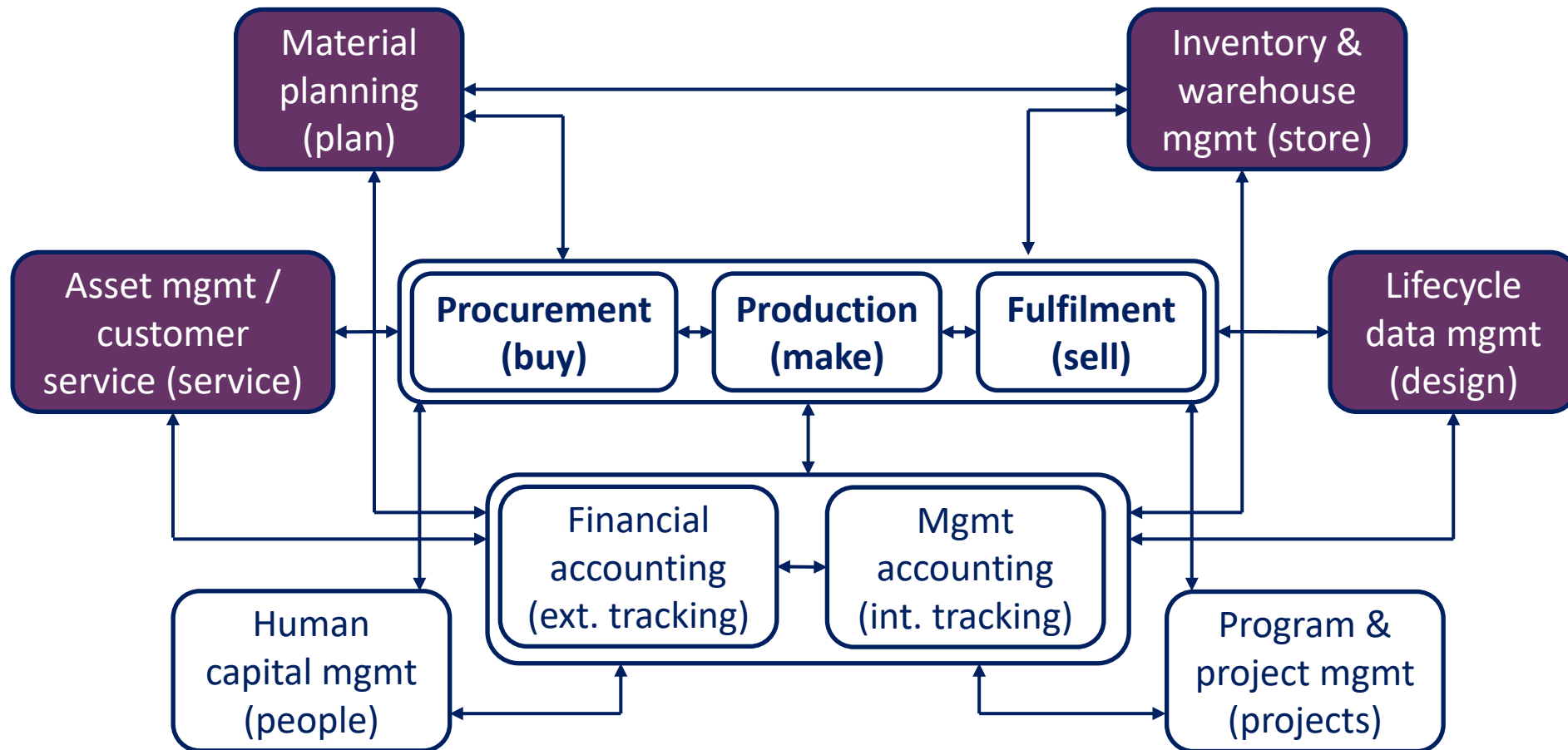
- *Preparing breakfast*
- *Placing an order*
- *Developing a budget*
- *Writing a work order*
- *Triaging a patient*
- *Cleaning a room*
- *Changing oil in a car*
- *Strapping down a truck*
- *Refuelling an aircraft*
- *Putting gas in the car*



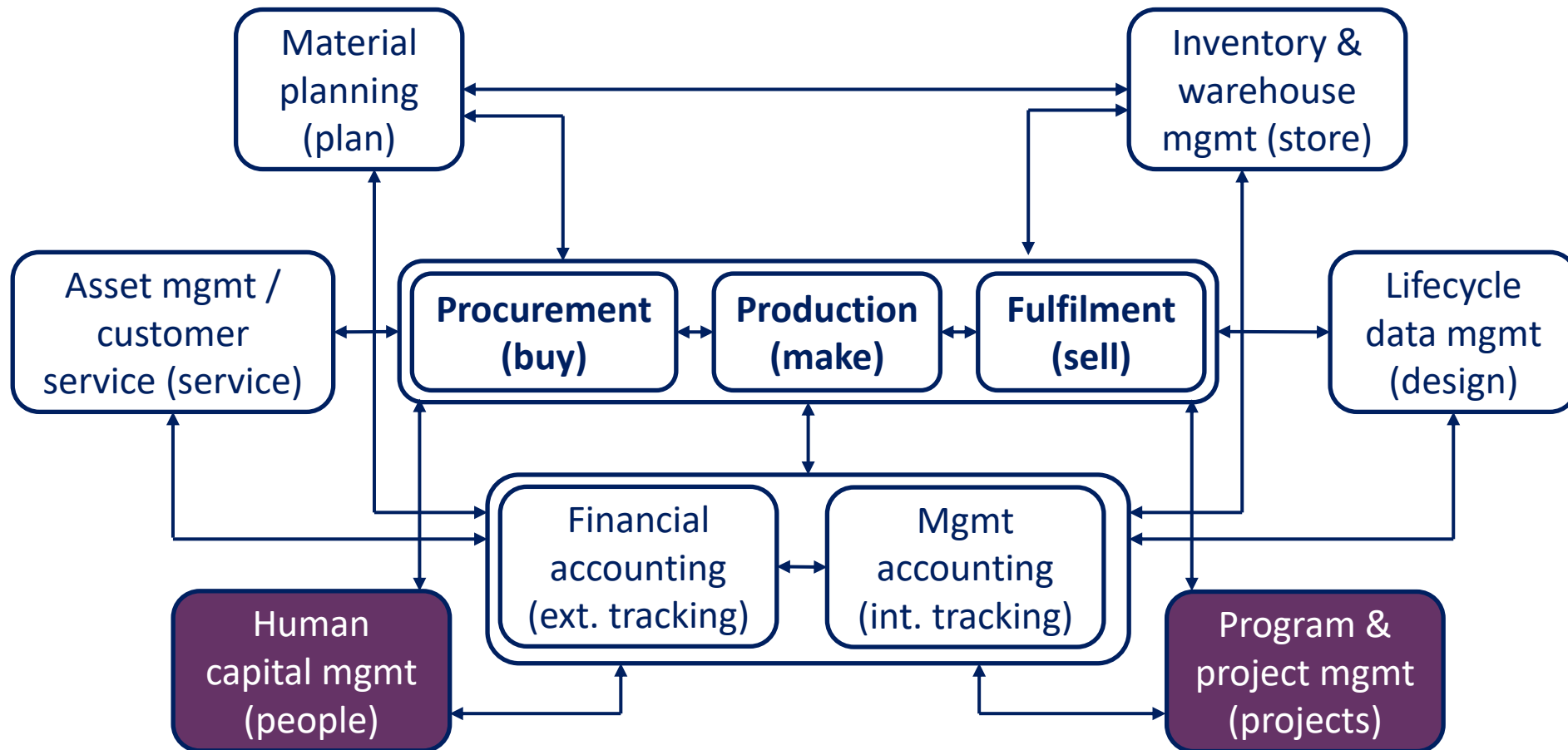
Business processes / 1



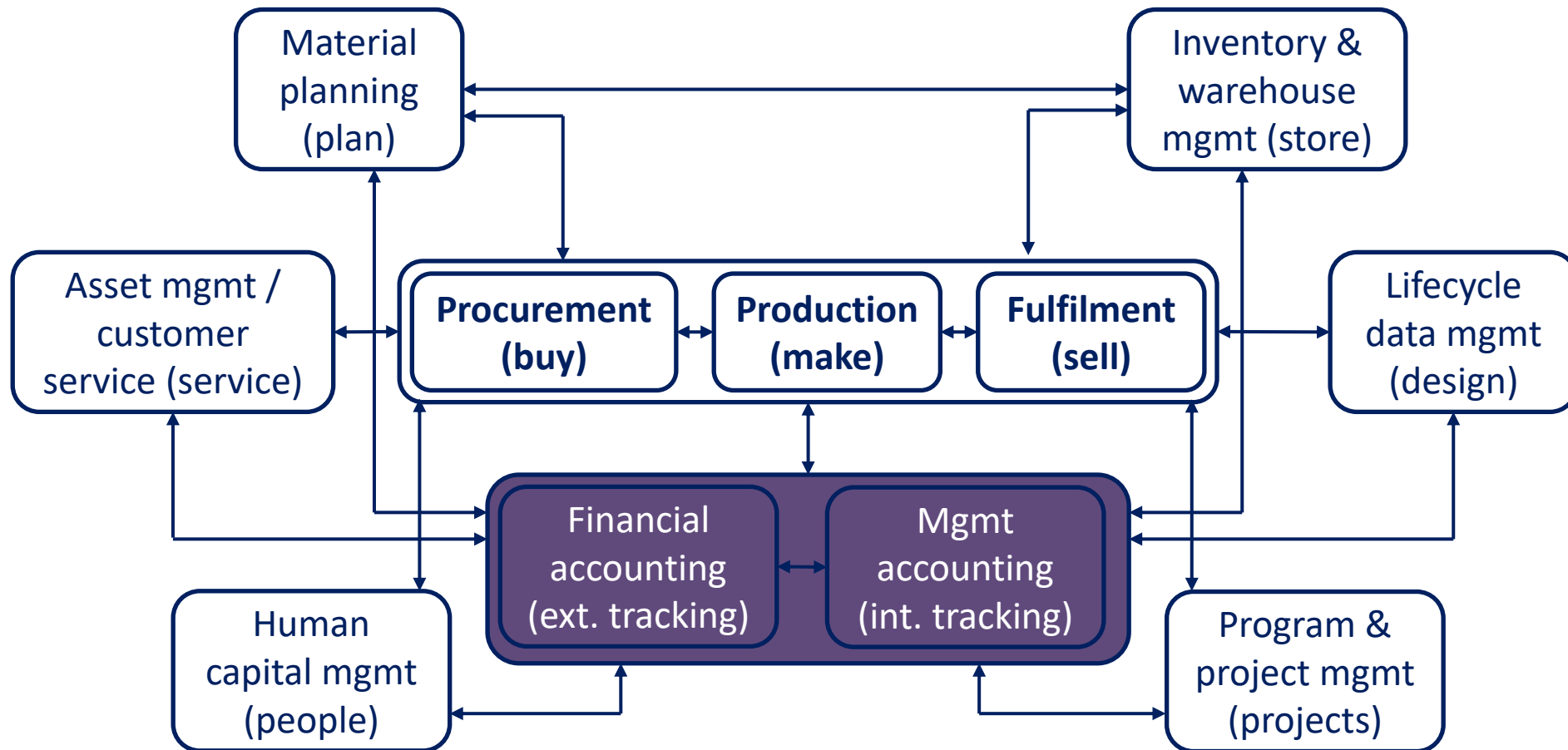
Business processes / 2



Business processes / 3



Business processes / 4



Business Process Oriented

The business process-oriented structure in an organization lead to

- **Cross-functional cooperation** in stead of “silo”

Which in turn lead to:

- **IT-based integration** of processes and applications → ERP systems
- Broader understanding of Logistics → not only transport anymore



Definition of new terms: “Logistics” ..

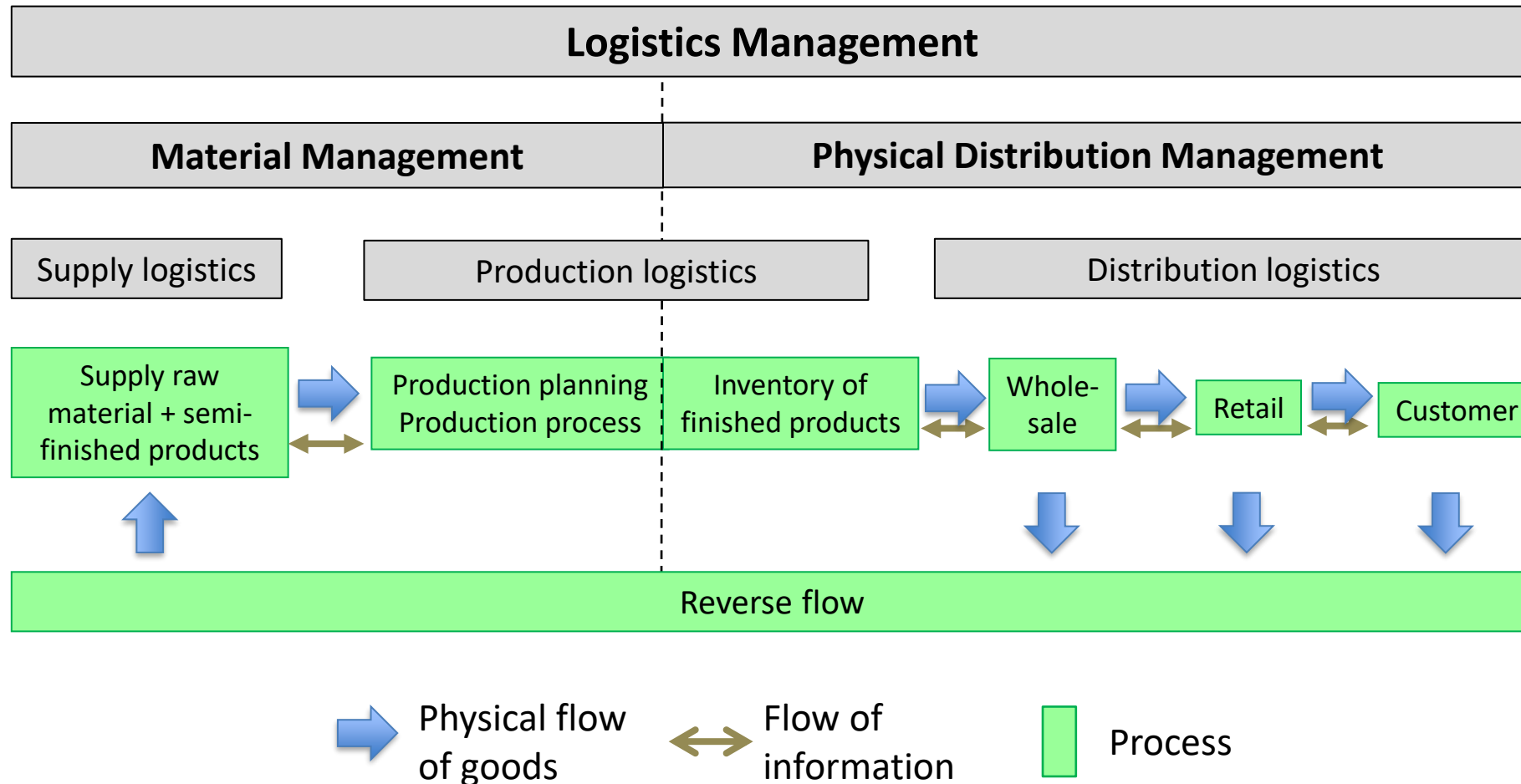
It started within the military.

Nowadays, Logistics entails the **organization, planning, control** and **execution** of the **flow of goods** from development and purchasing, through manufacturing and distribution to the consumer (end user), **up to and including the reverse flow**.



... and “Logistics Management”

LOGISTICS MANAGEMENT (= Business Logistics) is the generic term for all activities performed to control the incoming and outgoing flow of goods.



Globalisation & growing competition

Globalisation & growing competition again forced change, being shifts in:

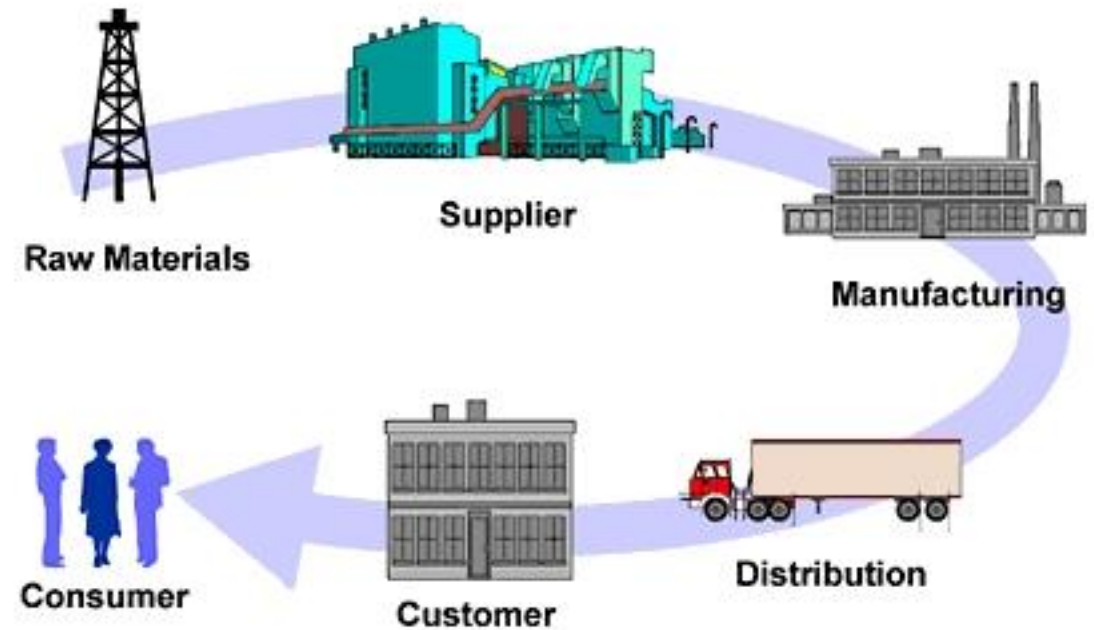
- Stock size (from “be-prepared” to “just-in-time/JIT”)
- Customer-orientation (from Henry Ford to “Know your customer”)

Which finally led to:

- **SCM** (Supply Chain Management)
- **CRM** (Customer Relationship Management)
- **ERP** (Enterprise Resource Planning)

Supply Chain Management

The chain is formed by the "flow of material" (material, but information and services as well) from the supplier of a resources via production, fabrication, assembly and warehouses to the final customer.



Customer Relationship Management (CRM)

CRM is the management of all those processes which give support for identification, selection, winning and binding of (potential) clients. ← “Know your customer”

Components of CRM:

- Functions: collect, store, distribute & analyse information
- **Operational CRM** (front-office CRM): support, help-desk and direct sales activities
- **Analytical CRM** (back-office CRM): marketing management, marketing campaigns

Data, data and more data

Next step is brought forth through:

- Growing possibilities in IT: cheaper, smaller, faster
- Growing amount of data already stored

Which made companies **start to making smart use of data everywhere**



Online Transaction Processing (OLTP)

- Ongoing
- Transaction oriented systems for supporting operational processes
- Making use of operational databases

Create Sales Order: Initial Screen

Create with Reference Sales Item overview Ordering party

Order Type Standard Order

1

2

Organizational Data

Sales Organization	AD06
Distribution Channel	10
Division	00
Sales Office	
Sales Group	

Online Analytical Processing (OLAP)

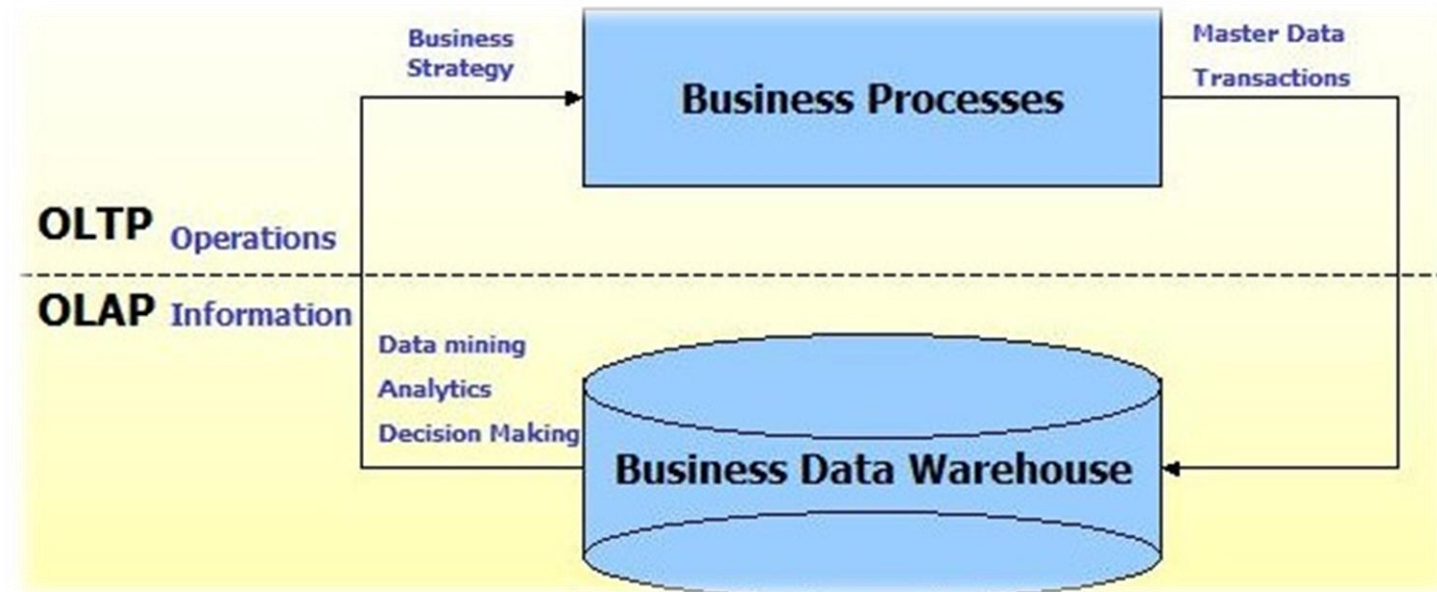
- Analytical systems for testing hypotheses, finding new relations, ..
= support for **decision making** on **strategic** and **tactical** level
- Making use of data warehouses / data lakes



Business Intelligence

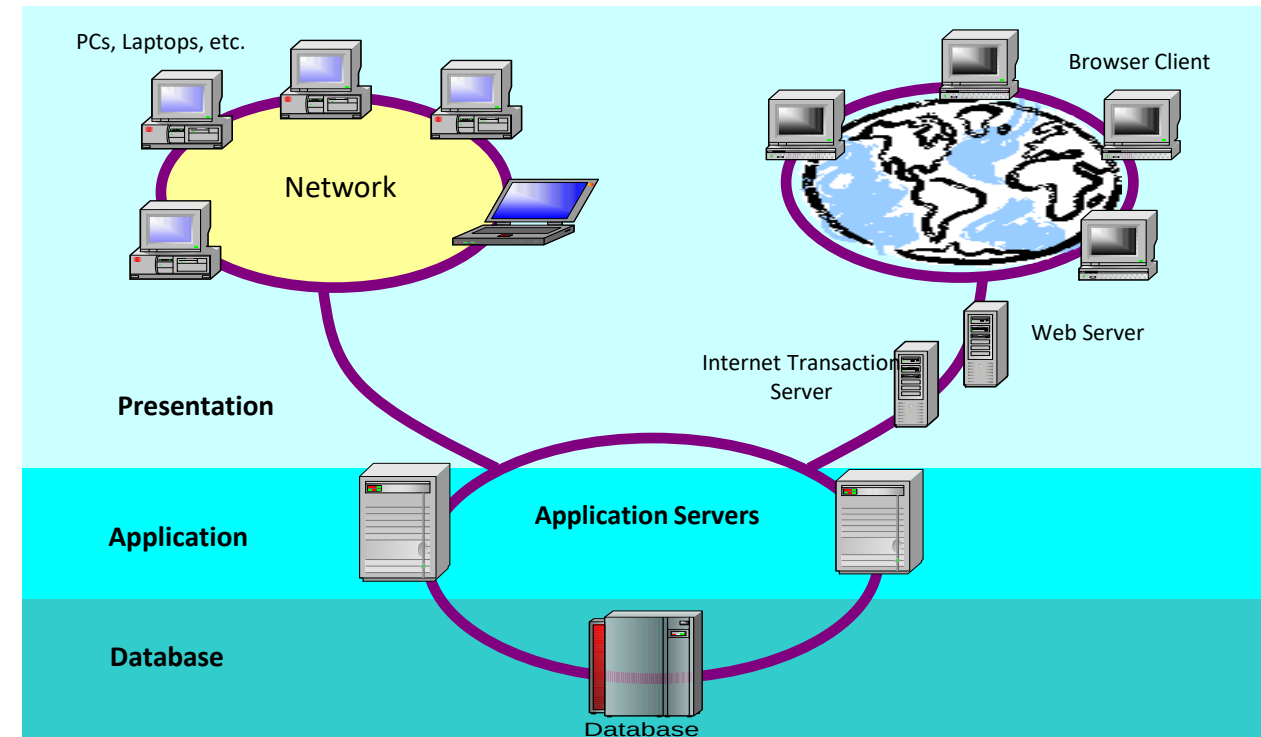
BI (Business Intelligence):

- OLTP + OLAP + data mining
- Decision support systems (DSS)
= support for making decision on ... + implementing / evaluating strategies
- Key Performance Indicators (KPI's) (using OLAP and data mining)



Architecture of ERP Systems

- Three-tier client-server architecture:
 - Presentation layer (client)
 - Application and data layer (server)
- Service-oriented architecture (SOA)
 - Communication via webservices/API's
 - Allowing for composite applications
- (Distributed) Relational Databases (more than 33.000 tables in SAP/R3 !!)
- Mostly modular structure



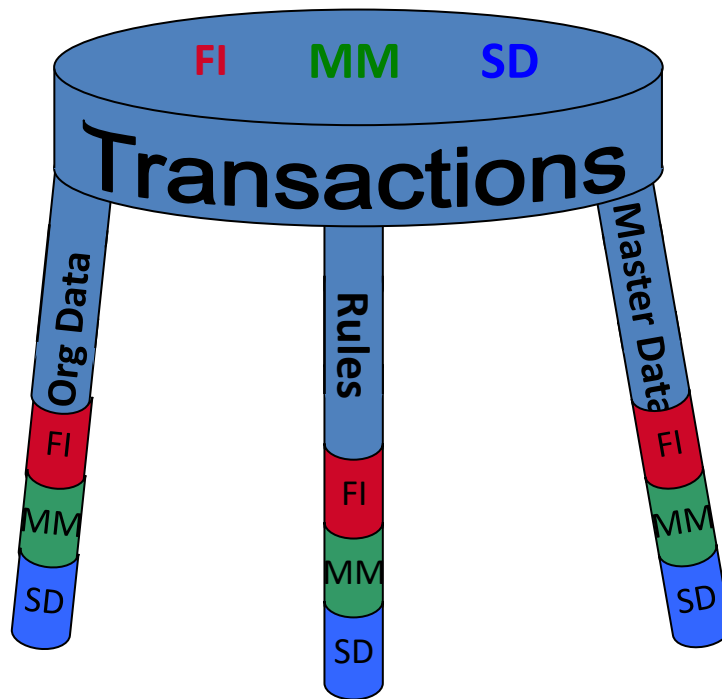
Bezos' API mandate (2002)

- 1) All teams will henceforth expose their data and functionality through service interfaces.
 - 2) Teams must communicate with each other through these interfaces.
 - 3) There will be no other form of interprocess communication allowed: no direct linking, no direct reads of another team's data store, no shared-memory model, no back-doors whatsoever. The only communication allowed is via service interface calls over the network.
 - 4) It doesn't matter what technology is used. HTTP, Corba, Pubsub, custom protocols — doesn't matter.
 - 5) All service interfaces, without exception, must be designed from the ground up to be externalizable. That is to say, the team must plan and design to be able to expose the interface to developers in the outside world. No exceptions.
 - 6) Anyone who doesn't do this will be fired.
- JEFF BEZOS

Basic Concepts in an ERP System: The Stool Metaphor

“The Stool”: a metaphor for the structure known from SAP

Four basic components needed to run & execute an ERP system:



- Transactions

- Organizational Data

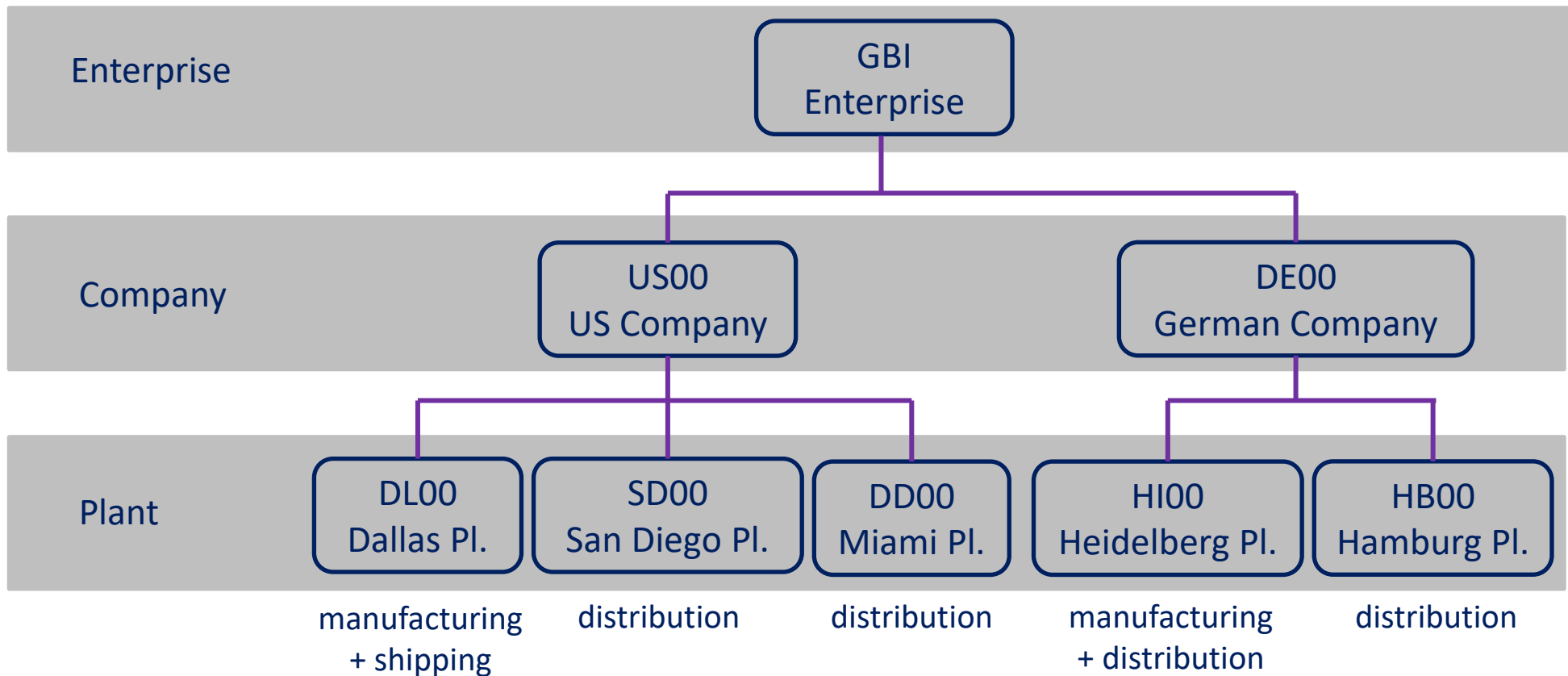
- Master Data

- Rules

- The legs which 'hold up' the transactions.
- Transactions cannot be run unless these are setup.
- The legs are typically configured during the implementation process.

Organizational Data

- Usually entered only once and “never” changed
- Representing the structure of an organization (own company)
 - .. reflect the hierarchy of the organizational units in an enterprise.
 - .. form the framework within which all business activities are executed



Master Data

- Representing entities associated with various processes
- Usually not changing so often
- Are created centrally and **shared among various business processes.**
- Contain **different views** on master objects from the perspective of different functions, e.g., production, warehousing, sales

Examples:

- Material master
- Customer master
- Vendor master



Rules

- Define which data changes are allowed and which not
- Are relatively fixed but may change due to policy changes
- Examples:
 - Only business customers can get credit, private customers must pay BEFORE delivery of products.
 - Products of category LUXURY can be sold with 20% off, the others only with maximumly 10% off.
 - Only accountants of plant X may enter orders for plant X.
 - How many fiscal periods are to be distinguished per year?

Transaction Data

= Data resulting from executing business process steps.

Examples of transaction data are:

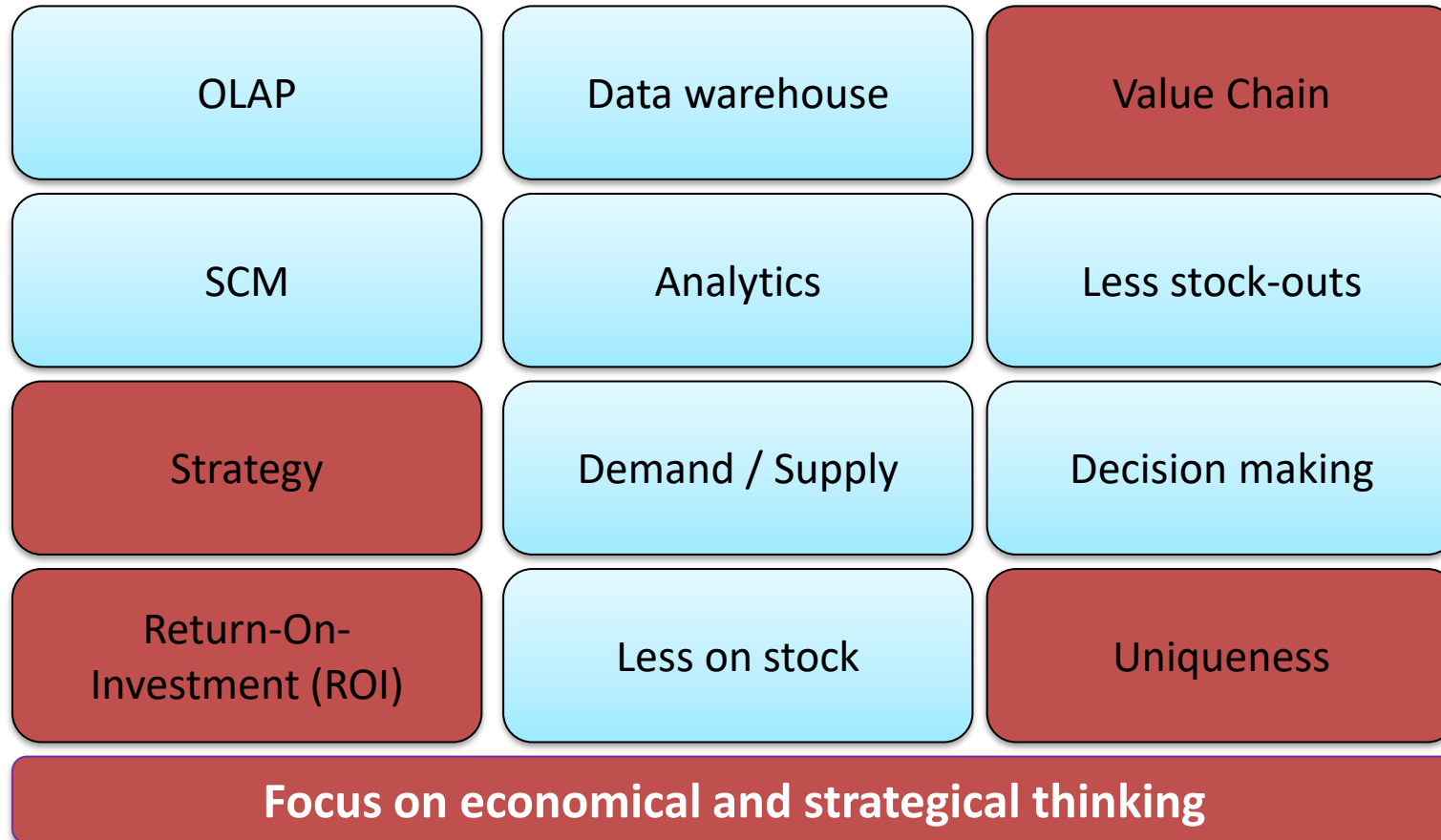
- The change in the number of a certain product stored in a warehouse, as result of a purchase order or a delivery into the warehouse
- The company is selling some material to one of the customers, so they prepare a sales order for the customer. The generation of a sales order is performing a transaction, the new sales order is transactional data: how many of what to whom when at what price

There are many info's and videos online on the difference of transaction and master data, e.g. <https://youtu.be/lv9P5D6yj30>

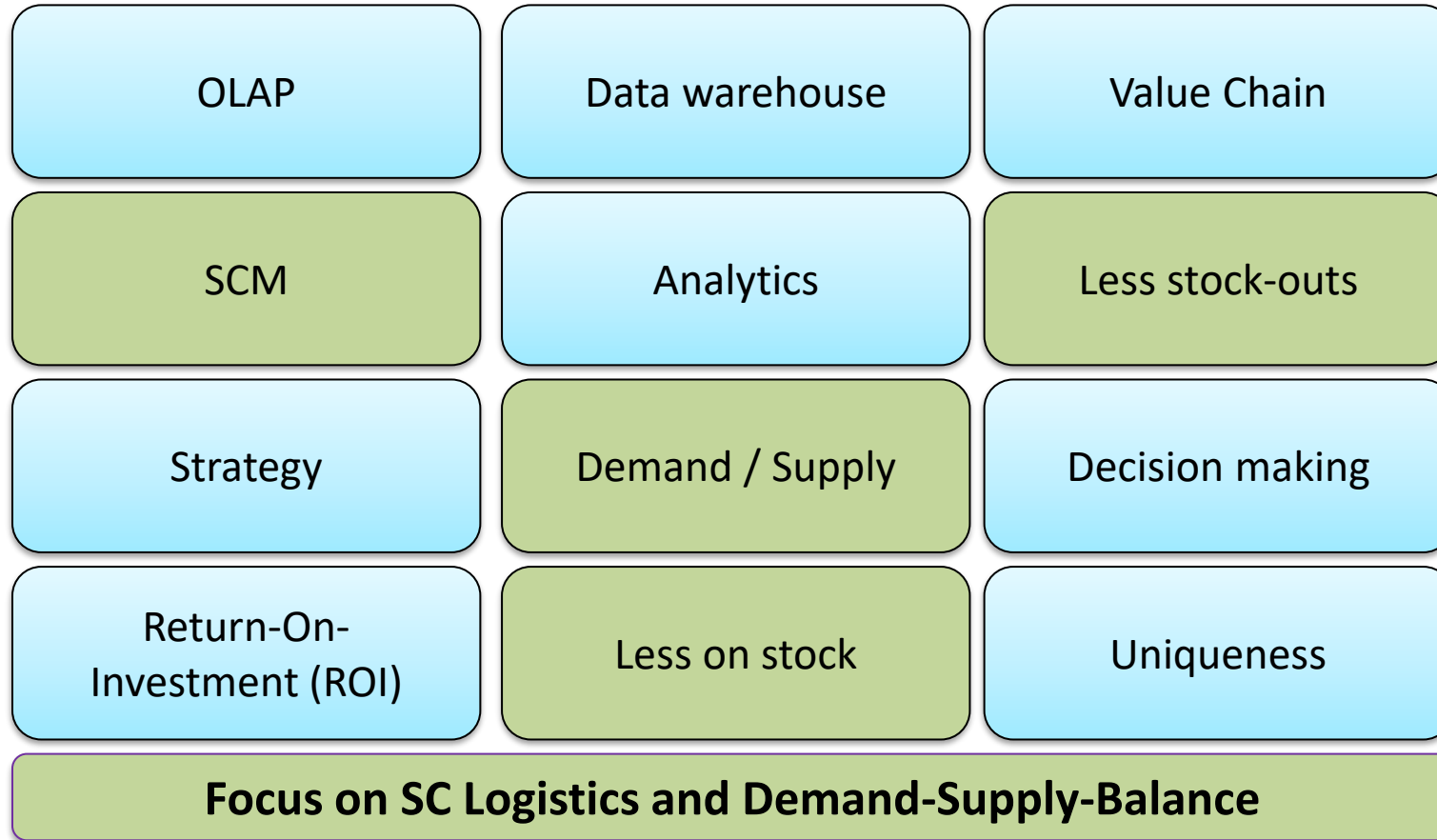
Many buzzwords so far

OLAP	Data warehouse	Value Chain
SCM	Analytics	Less stock-outs
Strategy	Demand / Supply	Decision making
Return-On-Investment (ROI)	Less on stock	Uniqueness

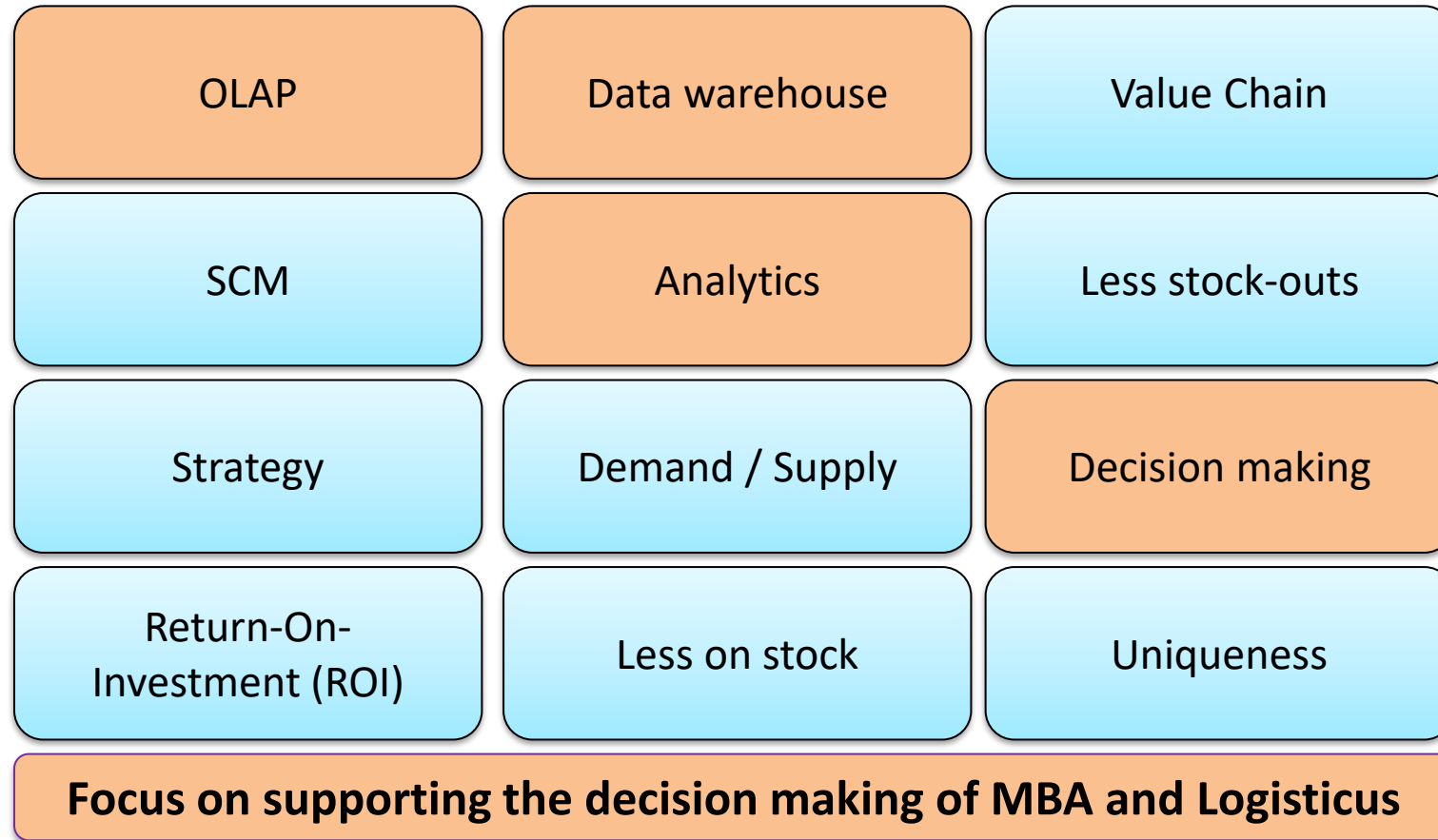
The MBA



The logisticus



The IT Business engineer



Coming up

- Major business processes
 - Procurement
 - Production
 - Warehousing
 - Fulfilment & Distribution
- Implementation of an ERP-system into an organization